

ELIO KOKA

SYSTEM INTEGRATION AND TEST ENGINEER

email: elio.koka@gmail.com | site: https://kokael.io | (248) 880-5549 | Redwood City, CA

PROFESSIONAL SUMMARY

Results-driven Systems Integration & Test Engineer specializing in robotics and autonomous systems. With over 3 years of experience in the end-to-end validation and commissioning of complex mechatronic platforms, proven leader in architecting test strategies and leading cross-functional debugging (HW/SW/FW) to ensure system reliability in production environments. Proficient in Python and C++ for automation and diagnostics, with deep expertise in system-level V&V and hands-on troubleshooting.

PROFESSIONAL EXPERIENCE

ROBOTICS TEST ENGINEER

Wheel.me | Apr 2024 - May 2025

- Spearheaded North American QA operations for autonomous guided vehicles (AGVs), leading cross-functional troubleshooting (EE, SW, Cloud) to **reduce system downtime by 20%**.
- Diagnosed and resolved complex system integration issues across hardware, firmware, and software, including CAN bus communication failures and power management systems (BMS/ECUs).
- **Engineered and maintained Hardware-in-the-Loop (HIL) test benches** for validating integrated electro-mechanical systems and rapidly adapted validation protocols for new component integrations.
- Developed automated testing pipelines using Python and Azure, **reducing manual validation efforts by 33%** and accelerating development and deployment cycles.
- Analyzed fleet operational data and test results to identify systemic failure modes, providing quantitative insights that guided priorities for software and firmware updates.

ROBOT BRING-UP ENGINEER

Carbon Robotics | June 2022 - Apr 2024

- Owned the end-to-end system integration, verification, and commissioning for a fleet of **70+ AI-powered autonomous agricultural robots** in a high-reliability production environment.
- Authored Python and Bash automation scripts for manufacturing test procedures, **slashing system verification time by 83% and saving over 1,500 engineering hours annually**.
- Led root-cause analysis for complex system failures across mechanical, electrical, and embedded software (C/C++) domains, utilizing diagnostic tools like **oscilloscopes, DMMs, and log analysis**.
- Managed integration and safety validation for **high-voltage systems** (up to 22kV), ensuring component compatibility and system integrity against rigorous standards.
- Introduced a Jira-based workflow to manage requirements and orchestrated collaboration between engineering teams (SW, EE, ME) to resolve integration blockers under tight deadlines.
- Provided critical **on-site deployment support**, rapidly resolving unforeseen integration issues and **training technicians on failure analysis techniques** using embedded Linux diagnostic tools.

CORE SKILLS & TOOLS

- **System Integration & V&V:** System Architecture Design, Verification & Validation (V&V), End-of-Line (EOL) Validation, Commissioning, System-Level Troubleshooting (HW/SW/FW), Hardware-in-the-Loop (HIL) Testing, FMEA, Root Cause Analysis
- **Programming & Automation:** Python, C++, Embedded C, Bash Scripting, Automated Testing Frameworks (Azure, Jenkins), CI/CD, Git
- **Hardware, Protocols & Diagnostics:** CAN Bus, High-Voltage Systems, Battery Management Systems (BMS), ECUs, Sensors (LIDAR, Computer Vision), Motors/actuators, Electro-Mechanical Systems, Diagnostic Tools (Oscilloscope, DMM, Logic Analyzer)
- **Domain Expertise & Methodologies:** Autonomous Systems, Mechatronics, Manufacturing Integration, Rapid Prototyping, Requirements Tracking & Translation, Cross-Functional Leadership, Documentation

EDUCATION

MICHIGAN STATE UNIVERSITY

Bachelor of Science, Computational Mathematics - Focus in Machine Learning
2017 - 2022